Module 05 Lab Exercise - Contribution Journal

ITAI 1371 - Introduction to Machine Learning

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My contribution to this lab focused on completing and verifying the data preparation steps in the Google Colab notebook. I worked through the notebook in order and completed the required coding tasks involving missing values, categorical data and feature scaling.

I began by running the initial dataset setup and examining the Titanic data for missing information. I used pandas to identify the number of missing values in each column. This allowed me to confirm that Age, Cabin and Embarked contained missing values and needed additional preparation before the data could be used.

For the missing values in the Age column, I calculated the median age and used the fillna() method to replace the missing values. After completing the imputation, I verified my work by checking the number of missing Age values again. The result was zero, confirming that the missing values had been successfully handled.

I also completed the categorical encoding portion of the lab. I used pd.get_dummies() to One-Hot Encode the Sex and Embarked columns. I used the drop_first=True option as required so that unnecessary duplicate information would not be created. I then reviewed the output and confirmed that the original categorical columns had been replaced by newly encoded columns such as Sex_male, Embarked_Q and Embarked_S.

Another part of my contribution was completing the feature-scaling task. I imported StandardScaler from sklearn.preprocessing, created a scaler and applied it to the Age and Fare columns. I reviewed the transformed output to make sure the scaling had been applied correctly.

In addition to the coding portion, I completed the knowledge check questions in the notebook. I explained why the median can be more appropriate than the mean when dealing with skewed data or outliers, described how One-Hot Encoding changes categorical data into a format that machine learning models can understand and explained why Decision Tree models generally do not require feature scaling. I also reviewed the notebook outputs throughout the process to make sure each required cell executed correctly. After completing the work, I prepared the notebook to be run from beginning to end so that the final PDF would clearly show the code, outputs and answers.

My contribution helped complete the preprocessing portion of the project and demonstrated the steps needed to transform raw data into a cleaner and more usable format for machine learning. Through my work, I was able to practice identifying data-quality problems, choosing appropriate preprocessing techniques, implementing those techniques in Python and verifying that each step worked correctly.

John Grubbs

I independently completed and ran my copy of the Module 5 Data Preparation lab. I loaded the Titanic dataset with Pandas and examined its structure and missing values. I identified missing data in columns such as `Age`, `Cabin`, and `Embarked`, then used the median of the `Age` column to replace its missing values and verified that no missing values remained in that column.

I used `pd.get_dummies()` to one-hot encode the `Sex` and `Embarked` columns while using `drop_first=True` to avoid unnecessary duplicate information. I then used `StandardScaler` from scikit-learn to standardize the `Age` and `Fare` columns. I reviewed the resulting DataFrame to confirm that the categorical columns had been converted into numerical features and that the selected numerical columns had been scaled.

I also completed the knowledge check in my own words. I explained why the median is useful for imputing skewed data, how one-hot encoding allows categorical information to be used by machine-learning models, and why decision trees generally do not require feature scaling. This assignment helped me practice the data-preparation steps needed before training a machine-learning model.